Paleomagnetism of Lower Cambrian volcanics and a cross-cutting Cambro-Ordovician diabase dyke from Buckingham (Quebec)

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The Buckingham area (Quebec) is characterized by Cambrian volcanic flows (573 ± 32 Ma) cut by a Cambro-Ordovician diabase dyke, which is dated at 497 ± 14 Ma by K–Ar method. The volcanics yield multicomponent magnetizations: a hard characteristic magnetization and a relatively soft one. These components could be separated easily and the intensity decay curves of the different components could also be obtained separately using Zijderveld's method. Three different groups of site-mean characteristic magnetizations can be distinguished, each representing a different stratigraphic level. The paleopoles derived from these magnetizations are, from oldest to youngest, 186°E, 10°S; 173°E, 01°N; and 154°E, 06°N. This is interpreted as a rapid motion of the paleopole for the North American craton during the Early Cambrian. The paleopole derived from the hard characteristic dyke direction is 123°E, 03°S. This magnetization direction was also found at one of the sites in the volcanics. This means that the contact test is positive and that the hard characteristic magnetization of the dyke is acquired at the time of intrusion.

La région de Buckingham (Québec) est caractérisée par des coulées volcaniques d'âge Cambrien (573 ± 32 Ma) coupées par un dyke de diabase d'âge Cambro-Ordovicien daté à 497 ± 14 Ma par la méthode K–Ar. Les roches volcaniques fournissent des aimantations à multicomposantes: une aimantation caractéristique dure et une autre relativement douce. Il est facile de distinguer ces deux composantes et les courbes d'affaiblissement de l'intensité de ces deux composantes différentes peuvent être obtenues séparément en utilisant la méthode Zijderveld. Trois groupes différents par la valeur moyenne des sites à aimantations caractéristiques peuvent être distingués, chacun représentant un niveau stratigraphique différent. Les paléopoles résultant de ces aimantations sont des plus anciens vers les plus récents, 186°E, 10°S; 173°E, 01°N; 154°E, 06°N. Ces résultats sont interprétés comme un déplacement rapide du paléopole du craton de l'Amérique du Nord pendant le Cambrien inférieur. Le paléopole mesuré sur le dyke caractérisé par une aimantation dure donne la direction 123°E, 03°S. Cette direction d'aimantation a été également observée à un des sites dans les roches volcaniques. Ceci signifie que l'expérience pour déterminer le contact présente des résultats acceptables et que l'aimantation caractéristique dure du dyke fut acquise au moment de l'intrusion.

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Introduction

From lower Paleozoic rocks of the North American craton, only a few well-documented paleomagnetic poles are available. Interpretation of these poles is hampered by the lack of accurate age determinations of the magnetizations from which the poles are derived, by the seeming complexity of the Cambrian apparent polar wander path (APW), and by the uncertainty about the integrity of North America at that time. Our knowledge of the APW during the Cambrian is far from complete. Most of the Cambrian poles plot in a broad band trending from the southwest Pacific to Japan. However, the sequence of poles within this band is not clear. There might be complications such as that proposed by Watts *et al.* (1980*a*,*b*) who found a loop in the APW based on the results from a section of sediments in Texas, an area that is generally considered to be an integral part of the craton.

Besides APW relative to the craton as a whole, relative motions of small blocks peripheral to the central craton of North America have been proposed and this could have affected the obtained paleopoles. For example, local rotations have been proposed by Gillett and Van Alstine (1979) for the Desert Range in Nevada, although one of the reasons for this rotation was that the pole they calculated did not fit on the assumed APW.

To be able to separate the effect of apparent polar wandering from the effect of relative plate motions it is evidently necessary to establish a reference APW path for the stable North American craton.

In order to increase the paleomagnetic data base for the Canadian part of the craton, Lower Cambrian volcanics (573 ± 32 Ma, J. Lafleur and D. D. Hogarth, in preparation) were collected near Buckingham, Quebec. These volcanics, which in fact are at the Precambrian–Cambrian boundary, are cut by a diabase dyke, one of numerous dykes of post-Grenville age (younger than 950 Ma), occurring north of the St. Lawrence graben. A number of these post-Grenville dykes have been sampled previously by Murthy (1971) for a paleomagnetic study. At that time, all the sampled dykes were believed to be of the same Late Precambrian age, but it is

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TABLE 1. Radiometric ages from rock suite related to the St. Lawrence rift

Location	Age (Ma)*		Method	Reference
Chatham Grenville stock	450	458	K–Ar	Doig and Barton (1968)
Rigaud stock	450	458	K–Ar	Doig and Barton (1968)
St. Sophie gabbro	520	529	K–Ar	Doig and Barton (1968)
Manitou Is. carbonatite	565	574	K–Ar	Gittins *et al.* (1967)
Lac Noir gabbro	560	569	K–Ar	Doig (1970)
Chicoutimi carbonatites	564	573	K–Ar	Doig and Barton (1968)
Mutton Bay syenite	568	577	K–Ar	Doig and Barton (1968)
Aillik Bay carbonatite	570	579	K–Ar	Doig and Barton (1968)
Sept-Iles anorthosite	540		Rb–Sr	Higgins and Doig (1977)
Buckingham latite	564	573	K–Ar	Lafleur and Hogarth (in preparation)
Radiometric ages from the dyke swarm†				
Gatineau Park	860±100		Rb–Sr	Bell and Blenkinsop (1977)
Gatineau Park	410±70		Rb–Sr	Bell and Blenkinsop (1977) Fahrig and Wanless (1963)
Rouge River (GSC 63-151)	790	800	K–Ar	Wanless *et al.* (1965)
Turbid Lake (GSC 66-119)	416±76	424±78	K–Ar	Wanless *et al.* (1968)
Fort Coulonge (GSC 65-113)	415±70	423±72	K–Ar	Wanless *et al.* (1967)
Gatineau–Lièvre region‡	446	454	K–Ar	Baird (1972)
Sudbury dike		438	K–Ar	Palmer *et al.* (1977)
Buckingham dike		497±14	K–Ar	This paper

*In the case of older K–Ar determinations both the age calculated with the old constant (left column) and the recalculated one (right column) are given (Dalrymple 1979).

†Selective dates; the whole of the swarm age interval is 900–400 Ma in the Ottawa Valley area.

‡Average of determinations from five locations.

now realized that the time span of dykes was several hundreds of millions of years. Murthy's (1971) results have to be reconsidered in that context. In the Buckingham area, one dyke was sampled at different locations for paleomagnetic and radiometric analysis.

Geology

In eastern Canada, numerous alkaline intrusive rocks and diabase dykes are associated with the St. Lawrence rift system (Table 1, Kumarapelli 1978). The age of the activity along the rift and the related rocks ranges from 1400 to 110 Ma (Doig 1970). Whereas the alkaline intrusive rocks closely follow the rift boundaries as defined by Kumarapelli (1978), the dykes form a swarm from Sudbury to Quebec City and possibly even beyond Sept Iles (Murthy 1971; Doig 1970). Within the interval of igneous activity along the St. Lawrence rift system, one episode is characterized by numerous alkaline intrusions having an age of 570 Ma (Table 1).

The Buckingham area is located about 45 km northeast of Ottawa (Fig. 1) near the north boundary of the Ottawa graben, an arm of the St. Lawrence rift system (Kumarapelli 1978; Doig 1970). According to Wilson (1916, 1920), a basement suite of metamorphic rocks including marble, quartzite, and biotite gneiss of Grenvillian age is intruded by Precambrian granite, pegmatite, and syenite and Late Precambrian kersantite and diabase. However, recent studies by Doyle (1969) and Lafleur (1978) indicate that Wilson's kersantites, immediately northeast of Buckingham (Fig. 2), are, in fact, thinly bedded latite flows. A K–Ar age of 573 ± 32 Ma is reported for these volcanic rocks by Lafleur and Hogarth (in preparation). Lafleur (1978) has distinguished two types of volcanic rocks: aphanitic lavas and fine-grained pyroxene biotite latites. A more detailed description of the rock type studied will be presented as a separate paper (Lafleur and Hogarth, in preparation). In the northeastern part of the area, the volcanic rocks are in direct contact with the Grenville rocks. In this area a dip of 26°W and a strike of 193° of the strata were measured. Whether this dip is primary or the result of later tectonic activity is not clear from observations in the field. Since minor block faulting is not uncommon in the whole Buckingham area (Lafleur 1978), it is possible that the dip of the volcanic flows is at least partly due to later tectonic activity.

The volcanic rocks that have been sampled are very dense, fine grained, and fresh. The volcanics are cut by

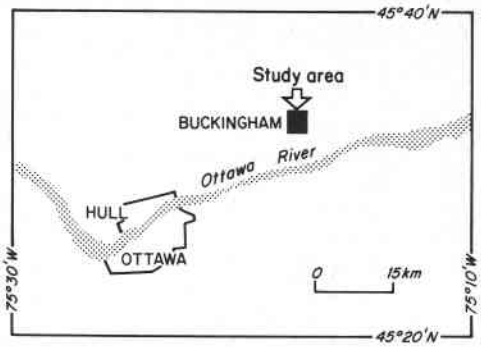


FIG. 1. Location of the study area.

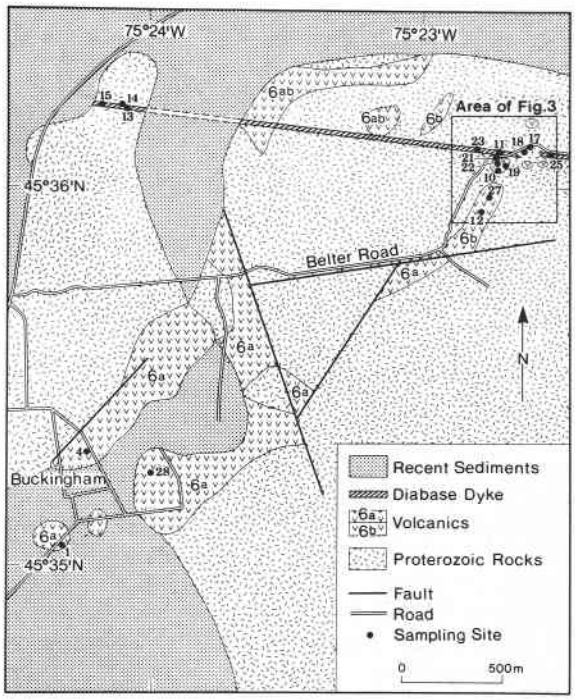


FIG. 2. Geological sketch map of the Buckingham area, after a more detailed map by Lafleur (1978).

an east–west diabase dyke, which dips vertically (Figs. 2, 3). This dyke is part of the east–west swarm that runs from Sudbury to Quebec City. The dates obtained from the dyke swarm range from 1400 to 400 Ma but in the Ottawa Valley area the range is from 900 to 400 Ma (Murthy 1971; Baird 1972; Palmer *et al.* 1977). The width of the dyke is approximately 20 m, although some slight variations occur. From Wilson's (1920) map it is clear that the dyke has a length of at least 8 km. Usually the dyke forms a topographic low. The contact between dyke and Grenville rock is locally intensively weathered because of three strong joint systems, which are more intense towards the margins. The orientation of one of the joint planes is vertical and parallel to the dyke

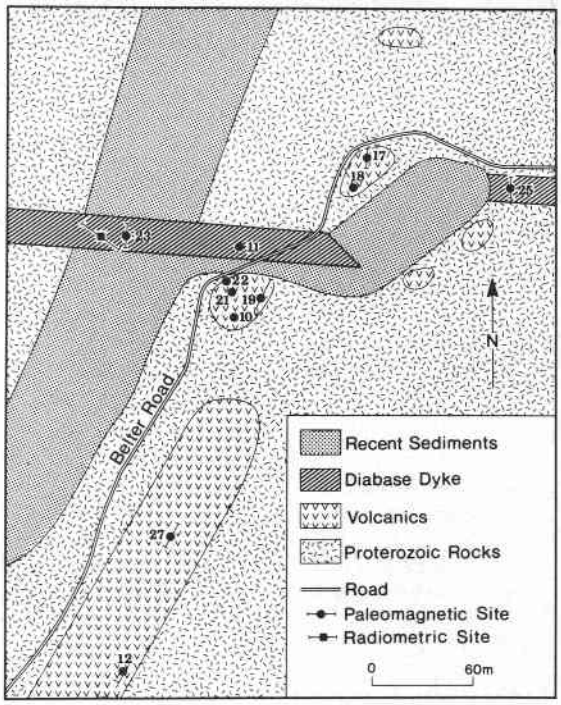


FIG. 3. Detailed geological sketch map of the area where the dyke cuts the volcanics.

direction. The other sets are perpendicular to the contact, with dips of about 75°E and 75°W. The dyke margins have a well-defined chilled zone, whereas the central zone is coarsely crystalline. At most places the rock is extremely hard and fresh. The cross-cutting relationship between the diabase dyke and the volcanics is clearly established in the field (Lafleur 1978). Radiometric age determinations have been made on samples of the dyke taken from the same localities as the paleomagnetic samples (Fig. 3) by A. Hayatsu from The University of Western Ontario (Earth Physics Branch contract No. OSU 79-281, January 1980) using standard K–Ar method (Hayatsu and Carmichael 1970). Samples for radiometric age determinations of the dyke were taken in two different areas approximately 2 km apart. One of the areas is close to paleomagnetic sampling site 23 (Fig. 3), the other in the area of paleomagnetic sites 13 and 14. At each radiometric sampling site, a number of samples were collected across the dyke and each of the samples was analysed individually. At site 14 no sign of weathering was observed, neither at the margin nor at the centre, neither in the field nor under the microscope. The K–Ar dates obtained from four different block samples from this area are very close together and give an average of 497 ± 14 Ma (Table 2). The situation at site 23 is less ideal. The joint system is well developed at this site and although macroscopically the rock looks very fresh, it

TABLE 2. K–Ar data

Site location	Sample No.	Distance away from contact zone (m)	K (90)	^{40}Ar (rad $\times 10^{-5}$ cm^3/g)	Calculated age (Ma)*
14	11	1.0	0.2478	0.5568	501
	12A†	1.5	0.2707	0.6042	498
	12B		0.2623	0.5792	494
	13	2.0	0.2729	0.5882	483
	14	10.0	0.2846	0.6533	511
Average					497±14
23	1	2.5	1.192	2.056	397
	2	2.5	1.462	2.504	394
	3	5.0	1.615	2.740	391
	4A†	7.5	1.657	2.535	356
	4B		2.081	3.232	360
	5A†	8.0	1.547	2.039	311
	5B		1.409	1.884	314

*Constant used: $\lambda\epsilon = 0.581 \times 10^{-10}$ year; $\lambda\beta = 4.962 \times 10^{-10}$ year; atomic $^{40}K/K = 1.167 \times 10^{-4}$ (Dalrymple 1979).

†Samples A and B are different pieces from the same block sample.

appears by microscopic examination that significant amounts of sericite are present. This is most likely to be the reason for the much younger radiometric age obtained for the samples from this site (Table 2). It is interesting to note that there appears to be a trend in the calculated age versus distance from the margin: the oldest ages are found in the margin and the youngest in the centre of the dyke. The discordance in calculated age between the two sites is not likely to be due to analytical errors as can be concluded from the close correlation of duplicate analysis from samples 4, 5, and 12. Nor can this discordance be explained by the presence of two separate dykes because although there is a discontinuity in the dyke between the two sampling areas there is no reason to believe that this gap where recent glacial deposits fill a topographic depression represents a major fault zone. Moreover it would be very coincidental that two dykes of different age would be exactly in line having the same width, the same outcrop phenomena, and the same petrological characteristics.

The problem we are signalizing here is a very serious one. It indicates that in general radiometric age determinations from dykes should be considered with caution, especially when the error is large. This means that radiometric dating methods put limitations on the use of paleomagnetic results even if the latter are excellent. An example of this, which is very similar to our case, can be found in Palmer *et al.* (1977) who obtained radiometric ages ranging from 401 to 492 Ma for samples from a dyke southeast of Sudbury.

For the time being we will only use the radiometric results from site 14 because the quality of the rock at this place is optimal and the error is small. More radiometric dating by other methods is in process and will be reported later.

Paleomagnetic sampling and measuring techniques

At 17 sites in the Buckingham area a total of 115 cores of 2.5 cm in diameter were drilled for paleomagnetic investigation (Figs. 2, 3). Orientation was by sun compass. Susceptibility of the specimen and of the rock at the outcrops was measured with a portable Bison susceptibility bridge (model 3101) equipped with an *in situ* coil. Remanent magnetizations were measured with a Schonstedt DSM-1. Thermal demagnetization was carried out in a Schonstedt furnace (TSD-1) in a residual field of 15 to 20 nT and in a larger furnace operating in residual field of <2 nT (Roy *et al.* 1969, 1972).

For the studies of alternating magnetic field demagnetization, results obtained with a three-axis tumbler (Roy *et al.* 1973) were compared with the results of a similar specimen demagnetized along three orthogonal axes, using the same demagnetization coil. For the collection of rocks studied here, the directions of the magnetization obtained by the tumbling method were the same as those obtained by the three-axis demagnetization method. The tumbling method was used for further alternating field experiments. A total of 132 specimens (76 from the volcanics and 52 from the dyke) were demagnetized and the results analysed using orthogonal projection diagrams (Zijderveld 1967).

Susceptibility

The *in situ* susceptibility of different rock types measured at the outcrops was compared with the volume susceptibility of specimens. Although the *in situ*

susceptibility was not always measured at exactly the same place where the cores were drilled, the agreement between *in situ* susceptibility and volume susceptibility is very good. The *in situ* susceptibilities of volcanics and dyke overlap so that *in situ* susceptibility could not be used for mapping.

In situ susceptibility values within most of the volcanics range from 1.88×10^{-2} to 9.40×10^{-2} S. Values below 1.88×10^{-2} S were found for some lower flows of the volcanic suite. *In situ* susceptibility values 10 times higher than normal were measured for the aphanitic lavas at site 28. Large variations occurring within one site indicate that the magnetic content of the volcanics varies rapidly even over short distances.

Volume susceptibility of specimens from the dyke ranges between 3.52×10^{-2} and 7.28×10^{-2} S. The lowest values were measured for specimens from the margins, consistent with the presence of finer magnetic grains. Variation of the *in situ* susceptibility within the central part of the dyke was related to the occurrence of joints and weathering. More intensive weathering resulted in increase of *in situ* susceptibility. It is interesting to see that the *in situ* susceptibility of the dyke at the place where it cuts the volcanics (Fig. 3) is 1.5 times higher than at other places. This must be attributed to the same source that caused the high potassium content in this part of the dyke (Table 2).

Natural remanent magnetization (NRM)

The volcanics show large differences in intensity, even within a site, and the NRM directions are distributed over the western hemisphere with positive as well as negative inclinations (Fig. 4). The within-site scatter is large for most of the sites. Only two out of eleven sites have an α_{95} smaller than 20° for the NRM directions. NRM intensities range from 0.1 to 40 A/m. Independently of the within-site precision the intensity range is always large. All specimens reveal high Q_n values.

The intensities and directions of the NRM of the dyke specimens are close together (Fig. 5). The highest NRM intensities of the dyke (1.0–3.0 A/m) are found in the centre. The finer grained rocks from the margin of the dyke reveal lower NRM intensities (0.7–1.0 A/m). The mean NRM direction for all the 48 dyke specimens is nearly vertical ($D = 002°$, $I = +86°$), the grouping is tight, and the error small ($\alpha_{95} = 4°$).

Characteristic magnetization of the volcanics

Thermal and alternating magnetic field demagnetization results from the volcanics agree very well. The remanent intensity decreases sharply between 560 and 570°C (Fig. 6), leaving no doubt that magnetite is the carrier of the characteristic magnetization. During thermal demagnetization the susceptibility remains the same, indicating that no significant chemical changes of the

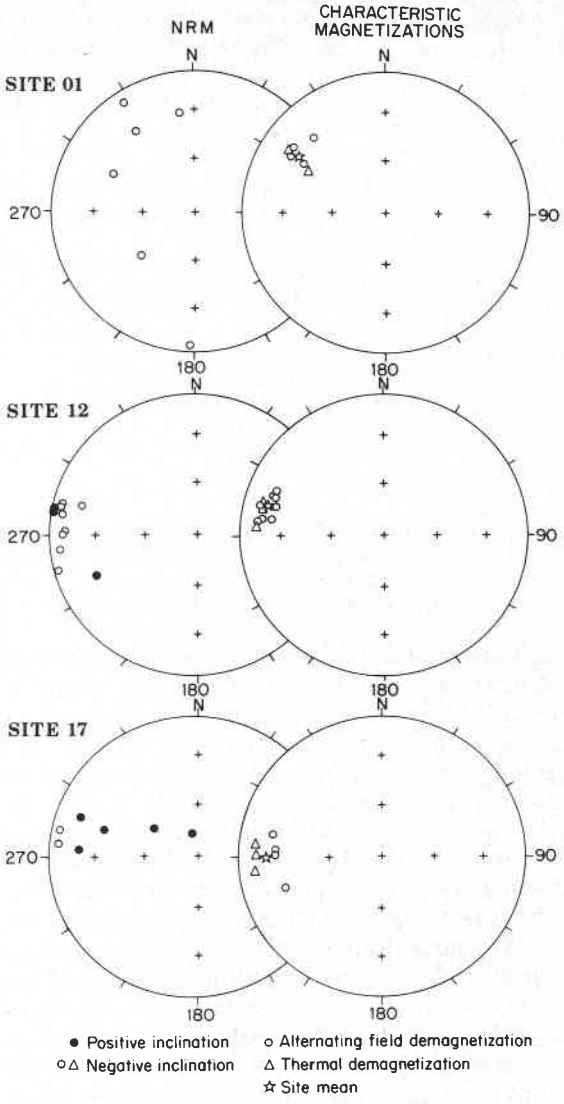


FIG. 4. Equal-area projection of NRM and characteristic magnetizations of some of the Buckingham volcanics.

magnetic phase take place. The intensity decay can therefore be attributed completely to unblocking of the magnetic moments of magnetite grains. For most of the specimens the characteristic direction is reached after heating to 525°C.

Alternating magnetic field demagnetization of the volcanics reveals two distinct magnetizations for each specimen (Fig. 7). Decay curves of the characteristic magnetizations were obtained separately from those of the soft magnetizations using the method described by Zijderveld (1975). This method has the advantage that intensity changes as well as changes in direction of the remanent vector can be readily seen. It also allows one to compare hardness spectra and to determine accurately

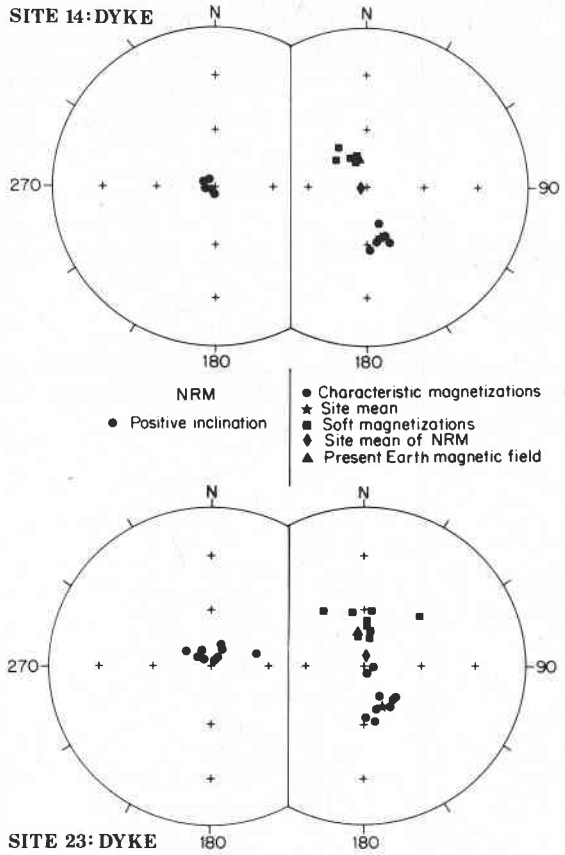


FIG. 5. Equal-area projection of NRM and characteristic magnetizations of two sites of the Buckingham dyke.

the median destructive field of each magnetization. This analysis shows that the soft magnetization is eliminated at a field of 40 mT. The characteristic magnetization in different rock specimens is removed by fields between 50 and 100 mT. Median destructive fields for different specimens range between 25 and 50 mT. These high values for the median destructive field and the sharp intensity decay close to the Curie temperature of magnetite (Fig. 6) indicate that the characteristic magnetization of the volcanic rocks is carried by single-domain grains of magnetite.

The effect of demagnetization on directions is illustrated in Fig. 4. The within-site grouping is very good (Table 3; Fig. 4) except for some sites for which only a few specimens give reliable results. The characteristic site directions from the volcanics as a whole (Fig. 8) are smeared from west to northwest. The lowermost flows of the suite (sites 17, 18) reveal a characteristic magnetization significantly different at the 95% confidence level from those obtained from volcanics of type 6a (sites 01, 04, 28). The intermediate flows give site means with circles of confidence partly overlapping both the lower and the upper flows. Tilt correction applied to groups I and II does not improve the grouping of the mean site directions significantly (Fig. 9). It is therefore expected that the difference in direction between site means most likely reflects a real difference in direction of the ancient geomagnetic field relative to the locality sampled. Therefore, a mean direction for each of the groups of volcanic flows has been calculated rather than an overall mean for all the volcanic sites.

Intermediate and hard characteristic dyke directions

From alternating magnetic field demagnetization diagrams two distinctive magnetizations could be derived for most of the dyke specimens (Fig. 10). The directions of both an intermediate and a hard characteristic dyke magnetization are tightly grouped (Fig. 5). The separation of the two magnetizations was less effective by thermal demagnetization than it was by alternating field demagnetization, probably because unblocking temperature spectra of the two magnetizations overlap more than the coercivity spectra. It was therefore decided to demagnetize the remaining dyke specimens with only alternating magnetic fields. The alternating field demagnetization pattern is approximately the same for all dyke specimens. Specimens from the centre of the dyke (e.g., 143A, Fig. 10) do not give significantly different demagnetization results than specimens from the margins (e.g., 237A, Fig. 10). A very soft temporary magnetization, demagnetized with fields of 2.5 mT, is most likely to be of viscous origin and acquired since the rocks were collected. A soft intermediate component is demagnetized with alternating fields between 2.5 and 10 mT. This component is approximately along the direction of the present local Earth magnetic field (Fig. 5). Slight deviations might be caused by the presence of temporary magnetizations. The hard characteristic magnetization is unblocked in fields between 10 and 40 mT (Fig. 10). The intensities of the intermediate and the characteristic magnetization are of the same order of magnitude. Usually there is only a small overlap of alternating field unblocking spectra of the intermediate magnetization and of the characteristic magnetization.

The hard characteristic magnetization has a southeast declination and a fairly steep downward inclination (Fig. 10). In some specimens the overlap of the two characteristic magnetizations was too large to allow separation. Also, some specimens did have a straight trajectory in the Zijderveld diagram and no separation of magnetizations was obtained because the direction was between the present Earth field and hard characteristic direction (Fig. 4, site 23). A possible explanation for the latter behaviour is that at each demagnetization step equal amounts of soft and hard characteristic magnetizations were unblocked. The results in which resolution

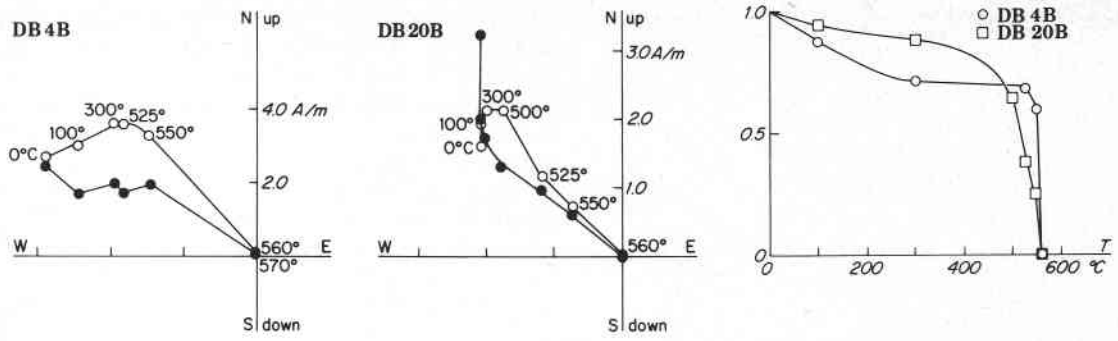


FIG. 6. Orthogonal projections (Zijderveld 1967) of the successive end points of thermal demagnetization results and normalized thermal intensity decay curves of some specimens from the Buckingham volcanics. Solid symbols indicate projection on the horizontal plane whereas open symbols indicate projection on a vertical plane.

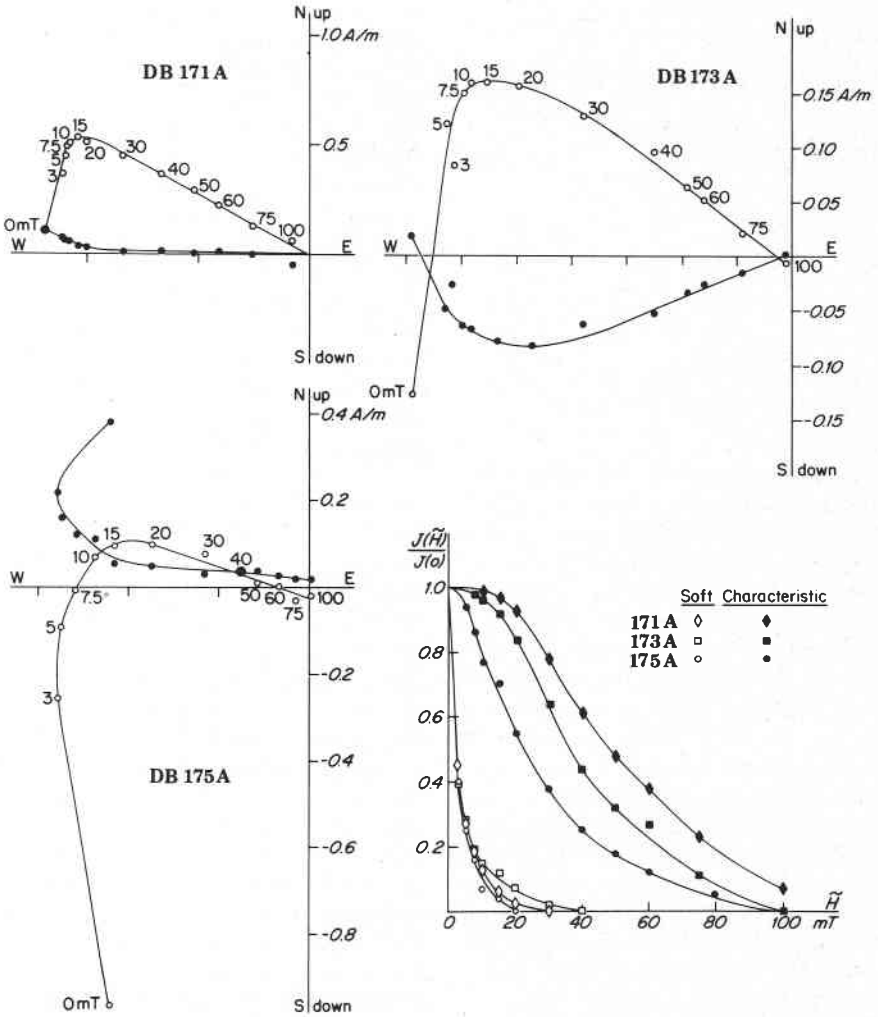


FIG. 7. Orthogonal projections (Zijderveld 1967) of the successive end points of alternating magnetic field demagnetization results and separate normalized alternating magnetic field intensity decay curves for soft and characteristic magnetizations (Zijderveld 1975) for some specimens from the Buckingham volcanics. Symbols as in Fig. 7.

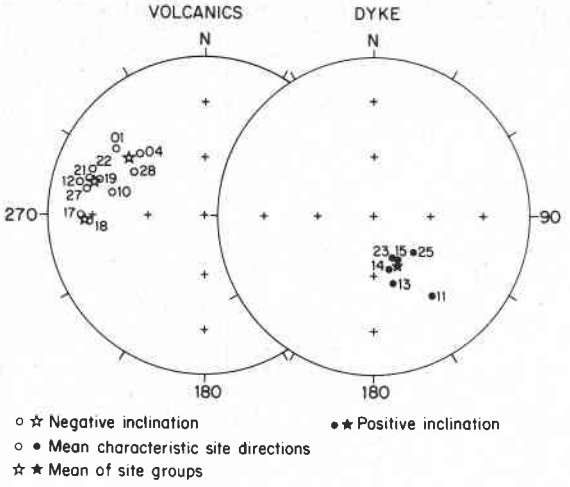


FIG. 8. Equal-area projections of characteristic site mean directions of Buckingham volcanics (before tectonic correction) and dyke.

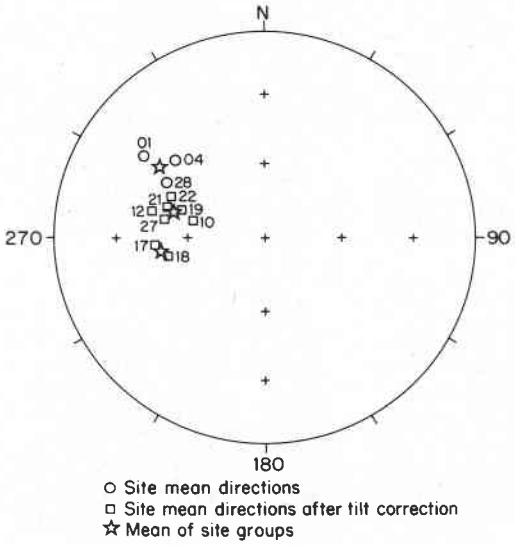


FIG. 9. Equal-area projection of characteristic site mean directions of Buckingham volcanics after tectonic correction for groups I and II.

of magnetization was not possible have not been used in the analysis.

The behaviour of the specimens from site 11, located where the dyke cuts the volcanics (Fig. 3), is quite different. Magnetizations that are more or less the same as at other dyke sites occur but they have a significantly lower inclination. A possible explanation for the difference in direction is that near the volcanics the dyke became geochemically contaminated, producing different magnetic minerals that became magnetized in a field deflected from the ambient field by the strong magnetization of the dyke or of the strongly magnetic volcanics. High susceptibility values and high potassium content near this site indicate that the magnetic and geochemical behaviour is different from that of the rest of the dyke. For these reasons site 11 is not taken into consideration in calculating the mean hard characteristic direction of the dyke.

Site means and the overall mean are given in Fig. 8 and Table 3.

Contact test between dyke and volcanics

A positive contact test between the dyke and the volcanics was found at site 22 (Fig. 11) where the volcanics are exposed within 10 m of the dyke contact. Intensive weathering of the contact between dyke and volcanics made it impossible to sample very close to the contact. The NRM's of three different cores from the volcanics have more or less the same direction as the characteristic dyke magnetization. Alternating field demagnetization diagrams show that a very large magnetically soft component is removed below 10 mT (Fig. 11). The direction of this component is the same as that of the hard component of the nearest dyke specimens. The inclination of these hard dyke components is shallower than the one of the rest of the dyke specimens as has been discussed in the previous section. The characteristic volcanic direction from the same specimens has unblocking fields between 30 and 75 mT. The intensity of the characteristic direction in the volcanics is only 2% of that of the characteristic direction in the dyke. The mean characteristic direction in the volcanics, calculated from the three specimens of this site, is $D = 291°$, $I = -24°$ ($\alpha_{95} = 20°$). There is a remaining magnetization after demagnetization at 75 mT that does not bear any relation to the characteristic direction in the dyke or in the volcanics. It probably originates from incomplete randomizing of the relatively very large unblocked magnetic moments at high, alternating fields. This would account for the large α_{95}. At site 22, the contact test is positive so that it can be concluded that the magnetization of the volcanics is older than the magnetization of the dyke and that the hard characteristic magnetization of the dyke was acquired during intrusion and is therefore of primary origin.

At another site in the volcanics close to the dyke contact (site 17, Fig. 3) an interesting observation was made. The intensity of the soft magnetization relative to the hard characteristic magnetization changes gradually throughout the site (Figs. 4, 7). The specimen closest to the dyke contact (171A) has the smallest soft magnetization and the specimen furthest away from the contact (175A) has the largest. The direction of the soft magnetization is close to the local present Earth magnetic field. Although the magnetic intensity is different, the median destructive field is about the same for all soft magnetizations (Fig. 7). The difference in intensity of

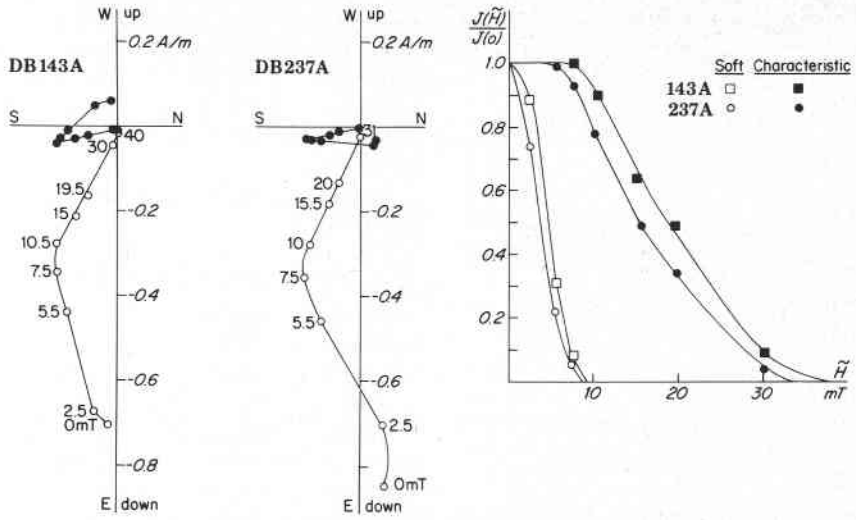


FIG. 10. Orthogonal projections (Zijderveld 1967) of the successive end points of alternating magnetic field demagnetization results and separate normalized alternating magnetic field intensity decay curves for soft and characteristic magnetism (Zijderveld 1975) for some specimens from the Buckingham dyke.

TABLE 3. Paleomagnetic results

Site No.	N_c sampled	$N_{c(s)}$* used	$D°$	$I°$	α_{95}†	Pole position	After tectonic correction: $D°$corr	$I°$corr	Pole position
Volcanics									
01	7	6(6)	305	−31	7	10°N 158°E			
04	5	4(6)	312	−43	12	7°N 147°E			
10	7	6(7)	283	−35	11	5°S 172°E	283	−61	24°S 152°E
12	7	7(11)	284	−19	5	2°N 178°E	283	−45	10°S 167°E
17	7	7(7)	270	−22	9	8°S 186°E	266	−48	23°S 176°E
18	5	3(3)	267	−27	13	12°S 187°E	260	−51	29°S 177°E
19	8	8(8)	289	−30	8	1°N 170°E	292	−55	12°S 155°E
21	6	4(4)	287	−24	13	3°N 173°E	288	−50	10°S 161°E
22	5	3(3)	291	−24	20	5°N 171°E	294	−50	7°S 157°E
27	5	4(4)	282	−25	18	1°S 187°E	281	−50	14°S 166°E
28	5	5(5)	301	−47	7	1°S 154°E			
Means									
Group III: 01, 04, 28			306	−40	14	6°N 154°E			
Group II: 10, 12, 19, 21, 22, 27			286	−27	6	1°N 173°E	287	−52	12°S 160°E
Group I: 17, 18			269	−25	13	10°S 186°E	263	−49	26°S 176°E
Dyke									
11	7	3(3)	143	+39	17	14°N 130°E			
13	8	5(5)	162	+56	14	6°N 120°E			
14	7	5(5)	165	+62	6	1°N 115°E			
15	8	4(5)	152	+66	13	7°S 123°E			
23	11	7(7)	156	+67	6	8°S 120°E			
25	7	6(6)	133	+64	9	10°S 136°E			
Mean			154	+63	7	3°S 123°E			

*N_c = number of cores; N_s = number of specimens.
†α_{95} is calculated giving core's unit weight.

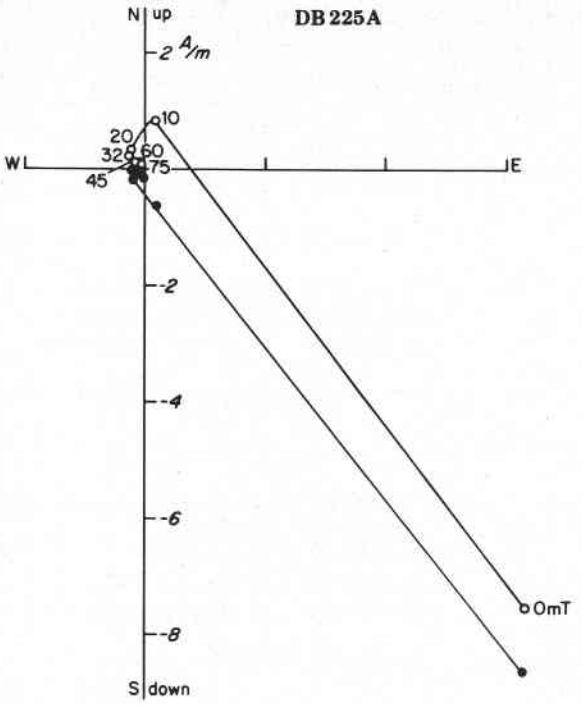


FIG. 11. Orthogonal projection (Zijderveld 1967) of the successive end points of alternating magnetic field demagnetization results of a volcanic sample from the contact with the dyke. The characteristic dyke direction is demagnetized with an alternating field of 10 mT and the characteristic volcanics direction is demagnetized between 40 and 75 mT.

the soft and the characteristic magnetizations for the various specimens is nicely reflected in the NRM directions, which are all approximately on a great circle passing through the characteristic volcanic direction and the present Earth magnetic field direction.

It is obvious that the difference in intensity of the soft magnetization cannot be related to thermal remagnetization during the process of dyke intrusion because the specimen that is most strongly affected is furthest away from the dyke. The difference in the magnetic hardness of the characteristic magnetization (Fig. 7) indicates that grain-size variations of the magnetite grains in the volcanics is probably responsible for it. It means that when these rocks were subjected to the same process they all responded in a different way. The rock with the coarsest grains would be most affected, resulting in the relatively largest soft component, which agrees with our observations. The nature of the remagnetizing process can be of thermal or chemical origin; even weathering cannot be excluded.

Discussion

The Cambrian paleomagnetic poles we consider most reliable are listed in Table 4 and plotted in Fig. 12. The numbers of the poles referred to in the text correspond to the numbers in Table 4 and Fig. 12. For the Buckingham poles, the positions before and after tectonic correction are given in Fig. 12. Although the dip of the lava flows is large, its origin is probably primary (D. D. Hogarth, personal communication, 1980). Therefore the poles without tectonic correction are preferred. The characteristic magnetization of the Buckingham volcanics is believed to have been acquired during initial cooling of the rock owing to a combination of the following observations: (1) its unblocking temperatures are limited to a narrow window immediately below the Curie point of magnetite (Fig. 6); (2) no other stable component of magnetization is present at temperatures higher than 580°C; (3) its resistance to alternating fields is high (Fig. 7).

The three Buckingham poles show the same kind of smearing as all the other Cambrian poles (Fig. 12). This can be the result of local rotations of small blocks or of rapid polar wander in Cambrian times. Local rotations can have taken place very easily in the Buckingham area because it is situated almost at the edge of the Ottawa graben. However, it is unlikely that relative rotation has occurred between the volcanics from which the Buckingham poles were obtained. The sites from which poles 1 and 2 were calculated were close to each other (Fig. 3) at each side of the dyke. There is some displacement of the dyke (Fig. 3) but the characteristic dyke direction at each side of the displacement is not significantly different (compare the means of site 23 and 25, Fig. 8). This means that if any relative rotation of the volcanics has occurred, it must have taken place between Early Cambrian and Ordovician. One possibility is that it happened at the time of dyke intrusion. In the field, however, no evidence for large-scale rotations could be found. At both localities the contact between Grenville gneiss and overlying volcanic flows was exposed. At the sites from which pole 1 was obtained, the lowest flows were sampled, whereas at the sites from which pole 2 was obtained, the samples were collected approximately 3 m above the contact. The difference between poles 1 and 2 can therefore be the result of apparent polar wander. The direction of APW is from pole 1 to pole 2. Although the stratigraphic relationship of the lava flows from which pole 3 was obtained relative to those of poles 1 and 2 cannot be established with certainty, the preferred interpretation is that the pole path goes from pole 1 through pole 2 to pole 3. This result indicates that the polar shift was fairly rapid at that time. The Buckingham poles give an age of 573 Ma, which is near the Cambrian–Precambrian boundary. This part of the path is in good agreement with the Early to Middle Cambrian path for North American poles that trends

TABLE 4. Paleomagnetic poles of Cambrian rocks from North America

Location	Age	Pole position	Reference
1. Lamotte Formation IIa	Late Cambrian	38°S 192°E	Al-Khafaji and Vincenz (1971)
2. Bonne Terre dolomite	Late Cambrian	38°S 190°E	Beales *et al.* (1974)
3. Lamotte Formation IIb	Late Cambrian	12°S 155°E	Al-Khafaji and Vincenz (1971)
4. Wood Canyon Formation (corrected by 36° rotation)	Early Cambrian	1°S 157°E	Gillett and Van Alstine (1979)
5. Colorado intrusives II	Late Precambrian – Early Ordovician	5°N 174°E	French *et al.* (1977)
6. Tapeats Formation	Early Cambrian	5°N 158°E	Elston and Bressler (1977)
7. Wilberns Formation	Late Cambrian	6°N 159°E	Van der Voo *et al.* (1976)
8. Carrara Formation group I (corrected by 36° rotation)	Middle Cambrian	7°N 158°E	Gillett and Van Alstine (1979)
9. Carrara Formation group II and Bonanza King Formation (lowermost member) (21, corrected by 36° rotation)	Middle Cambrian	9°N 160°E	Gillett and Van Alstine (1979)
10. Wichita granites (AF demagnetization)	Middle Cambrian (525 Ma)	13°N 147°E	Spall (1970)
11. Ophiolite Complex, Quebec	550 Ma	13°N 146°E	Seguin (1976)
12. Wichita granites (AF demagnetization)	Middle Cambrian (525 Ma)	16°N 148°E	Vincenz *et al.* (1975)
13. Bradore Formation	Early Cambrian	29°N 167°E	Rao and Deutsch (1976)
14. Wilberns Formation, Morgan Creek – Welge Members	Middle Late Cambrian (Early to Middle Franconian)	24°N 151°E	Watts *et al.* (1980*a*)
15. Riley Formation, Lion Mountain Member	Early Late Cambrian (Late Dresbachian)	27°N 146°E	Watts *et al.* (1980*a*)
16. Riley Formation, Cap Mountain Member	Early Late Cambrian (Early to Middle Dresbachian)	33°N 140°E	Watts *et al.* (1980*a*)
17. Riley Formation, Hickory Member	Middle Cambrian (Albertan)	34°N 145°E	Watts *et al.* (1980*a*)
18. Wichita granites (recalculated, Van der Voo *et al.* 1976)	Middle Cambrian	29°N 142°E	Ku *et al.* (1967)
19. Wichita granites	Middle Cambrian	30°N 138°E	Vincenz *et al.* (1975)
20. Carrara Formation, group I	Middle Cambrian	36°N 136°E	Gillett and Van Alstine (1979)
21. Carrara Formation, group II and lowermost member of Bonanza King Formation	Middle Cambrian	37°N 136°E	Gillett and Van Alstine (1979)
22. Wood Canyon Formation	Early Cambrian	38°N 134°E	Gillett and Van Alstine (1979)
23. Colorado intrusives	Cambrian–Ordovician	37°N 122°E	Larson and Mutschler (1971)
24. Rome Formation	Late Early Cambrian	38°N 144°E	Watts *et al.* (1980*b*)
25. Colorado intrusives I	Late Precambrian – Early Ordovician	15°N 142°E	French *et al.* (1977)
26. Colorado intrusives III	Late Precambrian – Early Ordovician	48°N 107°E	French *et al.* (1977)
27. Muav Formation	Middle Cambrian	55°N 110°E	Elston and Bressler (1977)
28. Abrigo Formation	Middle to Late Cambrian	59°N 89°E	Elston and Bressler (1977)
This study			
Buckingham volcanics			
Group I without tilt correction	573 Ma	10°S 186°E	
Group I after tilt correction		26°S 176°E	
Group II without tilt correction	573 Ma	1°N 173°E	
Group II after tilt correction		12°S 160°E	
Group III	573 Ma	6°N 154°E	

from 160°E 00°N to 130°E 40°N (Fig. 12). Poles given by Van Alstine and Gillett (1979) and also the pole of the Late Precambrian Johnnie Formation agree with the path for the Buckingham poles after correction for tilt of the flows. However, the dip of strata in Buckingham is most likely a primary one.

As said before, the results of the radiometric age dating should be considered with caution. Therefore, at this stage we will limit ourselves to comparing our paleomagnetic results only with other poles obtained from dykes of the same swarm and dated with the same radiometric method and not with other poles of the North

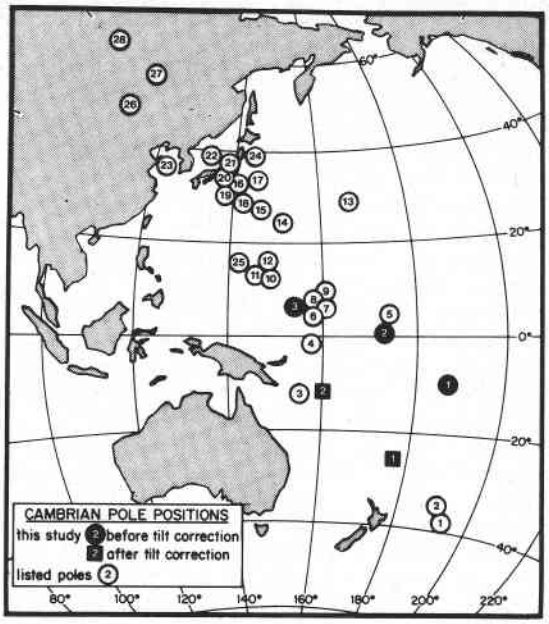


FIG. 12. Selected Cambrian poles for North America.

American craton. The pole position derived from the characteristic dyke direction is 123°E 3°S. This pole agrees reasonably well with the pole (140°E 5°S) obtained from a diabase dyke of approximately the same age (440 Ma, range: 401–492 Ma) from the Sudbury area by Palmer *et al.* (1977). Similar poles from the same dyke swarm were obtained by Murthy (1971), who then interpreted the age of the swarm as being Late Precambrian, although some ages around 450 Ma were obtained. It is now evident that the swarm is not a single event. If one looks at poles obtained from some of these dykes (e.g., GR 18, K–Ar = 423 ± 73 Ma, poles 100°E 32°S, Murthy 1971, pp. 802–805) one can suspect that more of these dykes that were interpreted as Late Precambrian are in fact early Paleozoic in age. Evidently one has to reconsider the poles obtained from this dyke swarm in relationship to radiometric data available. Since field relationship is not enough to distinguish the different dykes in this swarm (Palmer *et al.* 1977) one cannot assume or infer the age of a dyke by comparing with nearby dated dykes.

It is very interesting to note that three independent studies (Palmer *et al.* 1977; Murthy 1971; this paper) on Ordovician rocks from areas that belong without any doubt to cratonic North America yield poles and ages that are in good agreement with each other. These poles do not agree with other Ordovician paleopoles, mostly derived from sedimentary rocks from the Appalachians. Possible explanations for this discrepancy could be that rapid polar wandering occurred in the Ordovician that is not seen in the sediments, or that plate motion occurred between the two areas.

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